Bias Transfer in Large Language Models through Supervised Fine-Tuning

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19 July 2026; v4

Abstract

We examine how the bias of a training corpus transfers to a model fine-tuned on that corpus. We define bias to mean the degree to which the text departs from neutral, factual reporting and strays towards a subjective editorial writing style. Specifically, we fine-tune the Llama 3.2 3B Instruct model on four corpora that span a range of measured bias scores: NELA-GT (partisan US political news), NELA-GT-HB (a filtered subset of NELA-GT for high bias sources), NELA-PS (hyper-local auto-generated news content), and a corpus of Wikipedia articles on high schools in the US, which measured the least bias of the four. We evaluate the five resulting models (four fine-tuned conditions and the Llama Base) on 300 completions across 15 prompts each covering a salient topic (making $5\,\text{conditions} \times 300\,\text{completions} = 1,500$ completions total). Bias at the completions level is scored using an LLM-as-Judge rubric, and at the embedding-level with the Word Embedding Association Test (WEAT). Fine-tuning on the NELA-GT corpus raises mean output bias above the base model, with the largest bias shifts specifically on the topics of climate and immigration; filtering the NELA-GT dataset to only include the high-bias sources raises bias further (approximately $1.5\times$ the unfiltered corpus's shift over the base model), and also broadens the shift to topics that the unfiltered corpus left alone. Fine-tuning on the NELA-PS corpus reduces output bias. However, fine-tuning on the Wikipedia corpus produces the highest raw mean output bias, and the largest WEAT effect sizes, despite the corpus itself having the lowest bias scores. Wiki is the only condition that hallucinates at an abnormal rate. Correcting for the LLM-as-Judge's false positive rate, we calculate that 36.9% of its completions include fabricated legislation, policies or officials. This is in contrast to other conditions, none of which exceed 12.0% even raw, before correcting for false-positives. We attribute this paradoxical Wikipedia result primarily to hallucination inflation (when corrected, reverses the ranking). The effect is compounded by overfitting on a small training set, and amplification of pre-existing biases in the base model.

These results indicate that corpus-level bias scores are not reliable predictors of fine-tuned model bias, and that LLM-as-Judge evaluations that do not separate factual accuracy from biased framing will mark higher biases in models that produce more hallucinations.

1 Introduction

Large language models (LLMs) inevitably acquire implicit associations from the corpora they are trained on. Caliskan et al. [1] introduce the Word Embedding Association Test (WEAT), adapting the Implicit Association Test from social psychology to quantify associations between target and attribute word sets. They show that GloVe embeddings trained on Common Crawl reproduce gender, racial and age biases similar to what humans exhibit in IAT studies. This suggests that bias is not an artifact of pretraining or model architecture, but directly inherited from the training corpora, and is therefore measurable in the embedding space. So we reuse WEAT as a quantitative test of bias in our fine-tuned models, beyond superficial completions.

Performing supervised fine-tuning (SFT) on corpora specific to some domain is an increasingly popular route for adapting foundation models to specific use cases; however, it is not a neutral step.

Wang et al. [2] iteratively fine-tune a GPT-2 model on political news in the US and observe that across successive iterations, biases not only transfer but also are amplified into the model. Srivastava et al. [3] show that applying differential privacy to corpora used in fine-tuning limits bias amplification.

We extend this in three directions: 1. We use a modern instruction-tuned model (Llama 3.2 3B Instruct [4]) rather than GPT-2; 2. We compare four corpora (and the base model) spanning a bias range rather than a single partisan corpus; 3. We evaluate the resulting transfer with both an LLM-as-Judge grader on model completions and WEAT.

We measure bias in two ways: the first, which acts as our operational definition, is a LLM-as-Judge rubric score [5] (Section 2.2), ranging from 0–5 score for how strongly an article editorializes content, not of its specific ideological direction: on this scale, 0 denotes purely factual reporting, 3 a clear editorial stance, and 5 extreme partisan propaganda. This is now an established, standard methods for rubric-based evaluation at scale. From the score distributions, we analyze mean bias, bias rate, and uniformity. For a second evaluation of bias, we apply WEAT at the embeddings level.

The four original corpora span the range as follows: 1. NELA-GT [6], a collection of partisan news articles with a mean LLM grade of 1.818 / 5; 2. NELA-PS [7], a corpus of hyper-local news dominated by auto-generated statistics, with a mean score of 0.212 / 5; and 3. a Wikipedia [8] derived corpus of high school articles in the United States, scored at 0.060, serving as an essentially neutral baseline.

We also filter NELA-GT for high-bias articles to obtain a fourth corpus, NELA-GT-HB (High Bias). Only sources with a representative mean bias ≥ 3 / 5 are kept, which yields a final corpus mean of 3.300 / 5. Each corpus is used to fine-tune an identical Llama 3.2 3B Instruct [4] base model, via Low-Rank Adaptation (LoRA) [9]. By keeping architecture, hyperparameters and inference settings constant across all conditions, we ensure that behavioral differences could only be attributable to the training corpus.

Wikipedia is often treated as a maximally neutral corpus, and used as such for LLM training and evaluation, but superficial neutrality does not imply no bias. Navigli et al. [10] document that Wikipedia's editors are disproportionately male (87%) and English-speaking ($> 50\%$). This is directly relevant to Wiki, which is fine-tuned on the most neutral corpus by rubric score ($\mu = 0.060$). Nonetheless, it still produces the largest WEAT effect sizes and some of the most biased completions.

Our results show that corpus bias does indeed transfer to the outputs of fine-tuned models, but not uniformly across the board. Specifically, fine-tuning on partisan news shifts the overall bias distribution upwards, with the magnitude varying across topics. Filtered for high-bias articles, a model fine-tuned on such a corpus produces even more biased completions, and broadens this shift to topics that the model fine-tuned on the unfiltered corpus left unchanged. By contrast, the neutral Wiki condition produces a paradoxically large bias signal. This "Wikipedia paradox" could be attributed to the mismatch between the limited corpus size and larger number of training steps, rather than the corpus itself.

2 Methods

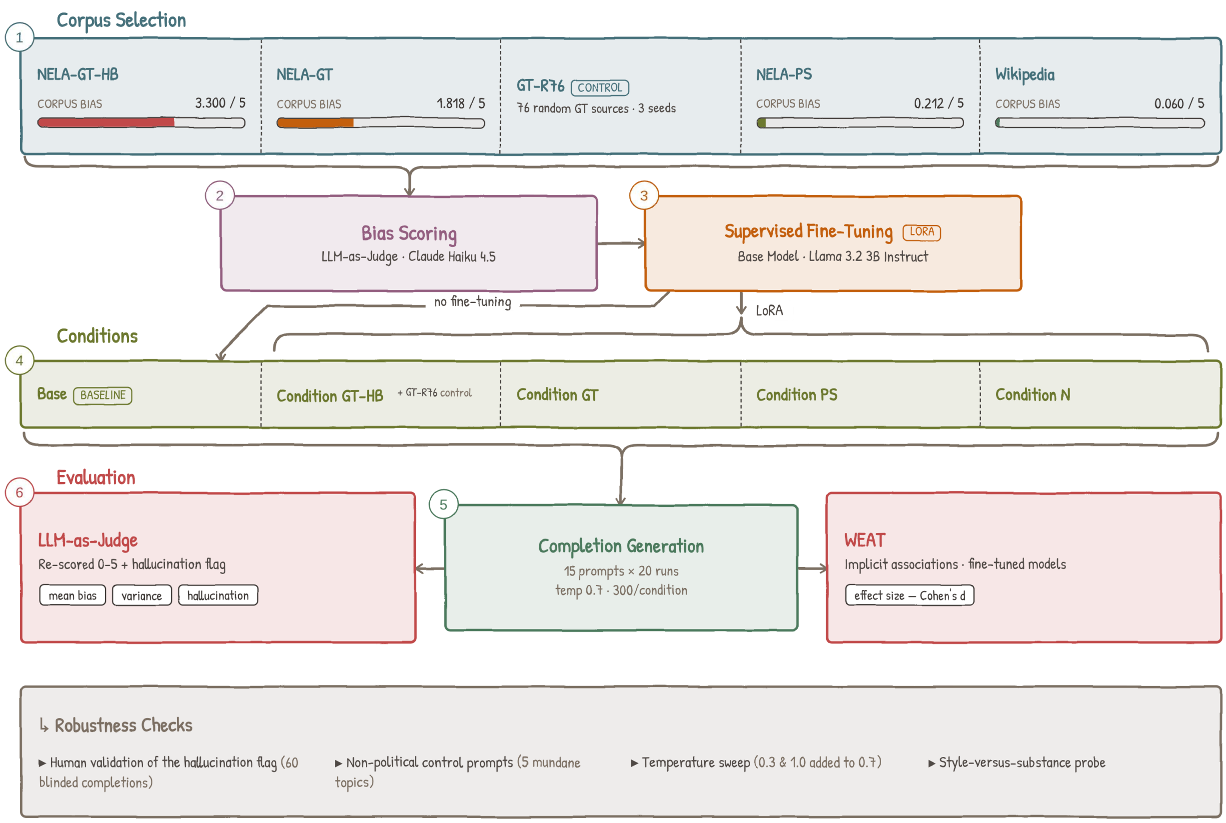

Figure 1: Pipeline overview.

2.1 Corpus Selection

We chose these corpora via three criteria. First, they had to consist of long-form article text also suited to the completion task the models are trained on, which rules out short, conversational sources such as Reddit or Twitter/X. Second, each had to admit a measurable amount of source-level bias, so the corpora could be ordered by bias: the NELA collections carry a partisan signal (via rubric grading and MBFC labels), and Wikipedia provides a near-neutral reference point. Third, each had to be publicly available and large enough to draw an identical training sample. However, it is notable that the four corpora are matched only on format and availability, not on their other properties: they differ in genre (partisan news, auto-generated local news, encyclopedia), authorship, size, and topic spread. Bias therefore co-varies with these factors, so the four-corpus comparison is best interpreted as a descriptive bias gradient rather than a specific manipulation of bias alone. In fact, the cleanest bias-isolating contrast is NELA-GT versus its filtered subset NELA-GT-HB (Section 2.1.2), which hold source set, sample size, training steps, and LoRA configuration fixed and vary only the bias of retained sources. We revisit this confounding in the Limitations.

2.1.1 NELA-GT

The NELA-GT corpus is sourced from the misinfo-general dataset [11], which takes a deduplicated aggregate of six NELA corpora [6] spanning from 2017 to 2022, compiling 4.2 million articles from 488 distinct publishers [11], of which 474 are present in our working corpus. Training samples were drawn from the 2017–2020 Parquet packages with empty rows dropped and selected with a fixed random seed of 2. 500,000 samples were taken and used as the training set for GT.

2.1.2 NELA-GT-HB

The NELA-GT-HB (High Bias) corpus is constructed using NELA-GT as a baseline. We first scan the original NELA-GT corpus, and enumerate the number of articles represented by each source. A representative sample of 20 articles is selected from each source with $\geq 1{,}000$ articles, with grades from the original NELA-GT corpus audit reused where available, and additional articles graded to reach 20 per source. Filtering at the source level is necessary for the LLM judge: using API calls to grade all 4.2 million articles individually would require three orders of magnitude more Haiku calls. A mean bias score is taken for each source, and all articles from sources ith ≥ 3 bias are chosen for the final corpus. This yields a total of 76 sources and 656,001 articles for NELA-GT-HB, of which 500,000 are randomly sampled for the training set via a deterministic hash. Checking each source threshold gives 1.39M articles for ≥ 2.0, 1.01M for ≥ 2.5, 656k for ≥ 3.0 and 284k for ≥ 3.5; we set $t = 3.0$ as the strictest threshold whose article pool still clears 500k comfortably. An independent validation of the final corpus confirms the high bias: as shown in Table 5, a 500 article sample scores a mean bias of 3.30, with 93.0% of articles having bias ≥ 2, compared to NELA-GT's 53.6%. We also use MBFC metadata as a cross-check only (it is publisher-level and US-centric, per the dataset README), which confirms the high-bias nature of the filtered set (as is shown in the source map Figure 2). Figure 10 shows an information fingerprint of all the sources considered; see Appendix G for the full source-selection figures, including the threshold choice (Figure 9).

2.1.3 NELA-PS

The NELA-PS corpus is a separate collection of 7.9 million articles from 1,093 sources, spanning March 2021 to January 2024 [7]. These sources are primarily dominated by Metric Media (6.99 million articles), a media network that produces articles with auto-generated local statistics that hold no editorial voice. This corpus is the training set for PS. Similar to GT, 500,000 samples were taken and used for training PS.

2.1.4 Wikipedia-derived

The Wikipedia corpus was built by recursively traversing a graph built on Wikipedia's category pages [8]. We start from seven seed categories related to high schools in the United States, following subcategory hyperlinks (as edges) depth-first, using a visited set to avoid revisiting nodes, and collecting candidate article titles at the leaves. Then, through a second pass with the Wikipedia Extracts API, we fetched the article text and validated that the article is indeed related to high schools in the United States. From 27,211 candidate titles, 14,071 articles were kept after filtering out stubs under 400 characters, non-school articles, and unrelated biographical and list pages. Of these, 10,000 were finally taken and used as the training set for Wiki. Sections irrelevant to bias, such as Alumni, References, See Also, etc. were stripped before saving. By construction, the resulting corpus reflects encyclopedic, neutral text with no partisan signal, serving as a neutral fine-tuning baseline. Figure 7 in the appendix reports per-token keyword frequencies across the school types represented in the corpus, indicating that the corpus is not topically uniform, despite aggregate neutrality.

2.2 LLM-as-Judge Pipeline

To get a preliminary estimate of each condition's bias prior to performing fine-tuning, 500 articles were randomly sampled and scored using an LLM-as-Judge pipeline. This pipeline automates the bias scoring process, using Claude Haiku 4.5 (claude-haiku-4-5-20251001) [12] as a judge. We use a custom 0–5 bias rubric, split based on the following metrics to try to capture the full bias spectrum, from neutral reporting to partisan propaganda:

- 0: purely factual, no discernible bias
- 1–2: subtle framing, barely detectable
- 3: moderate bias, clear editorial stance
- 4: strong and overt advocacy, misleading framing
- 5: extreme bias, disinformation, propaganda

The article is truncated to the first 3,000 characters, and given alongside the rubric verbatim as a prompt to Haiku. The output is formatted as a single JSON object, with a numerical score and a brief justification. Parse failures are assigned a score of -1 and skipped.

Corpus bias scores are reported in Table 5. The four corpora span a broad bias gradient, making them suitable for the intended experimental contrast of a high-bias partisan vs. its higher-bias filtered subset vs. low-bias auto-generated vs. near-neutral dataset.

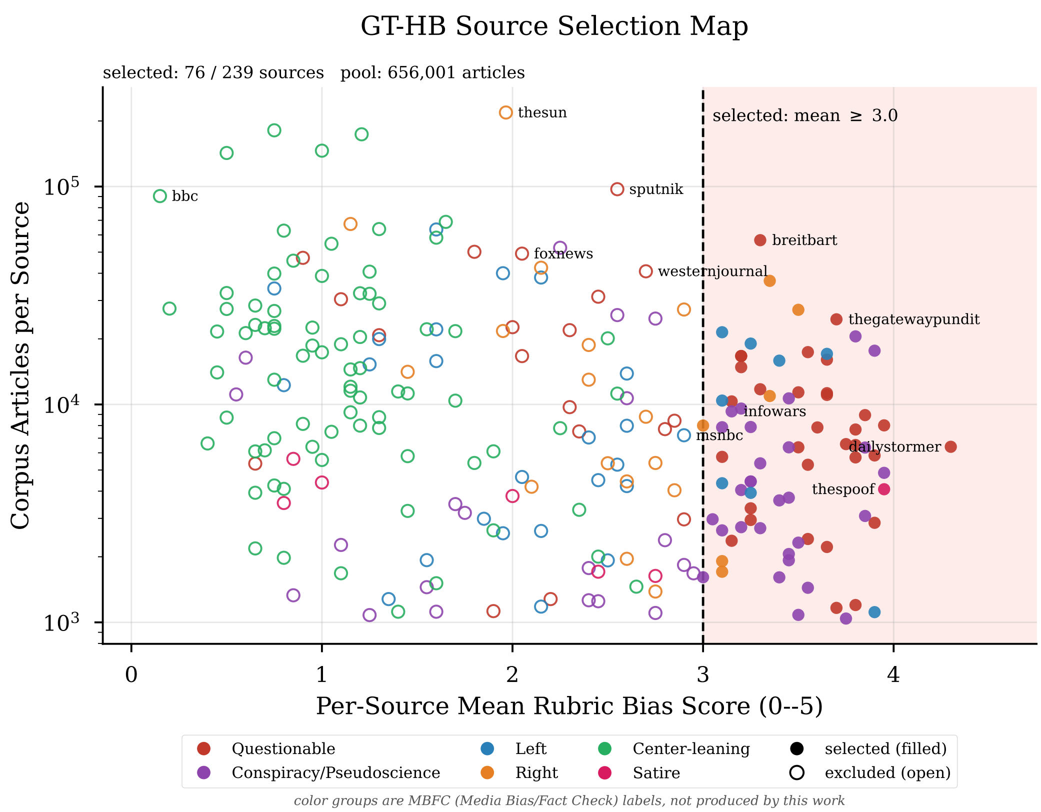

Figure 2: GT-HB source selection map. Each point is one audited NELA-GT source, plotting its per-source mean rubric score against its corpus article count (log scale). Filled markers are sources selected for the GT-HB whitelist (mean ≥ 3.0, vertical line); open markers are excluded; color gives the MBFC label group.

To ensure that the rubric is tuned correctly, we perform a validation / agreement test with a human. Taking a 30 article sample from each of NELA-GT and NELA-PS, a human annotator scored NELA-GT on the full 0–5 rubric used throughout and NELA-PS on a smaller, representative 0–3 rubric. Results are shown in Table 1.

Table 1: Human-LLM Agreement

Corpus	Exact Agreement	Within ±1	Pearson r	LLM Drift
GT	43% (13/30)	80% (24/30)	0.640	+0.367
PS	53% (16/30)	100%	—	+0.333

Note: Pearson r not reported for NELA-PS, due to Haiku's scores being almost degenerate: 87% score 0 in the 30 article validation. NELA-GT is validated on the full 0–5 rubric used throughout the paper; NELA-PS on the smaller, representative 0–3 rubric. Drift = mean(human) − mean(LLM)

To note is the Cohen's d [13] value for Haiku's scores (NELA-GT v. NELA-PS) is 1.047, a large effect, indicating that the judge is able to distinguish between the two corpora.

As another check that these scores are not an artifact of a unique model, we re-grade all 1,500 completions (5 conditions $\times$ 15 prompts $\times$ 20 completionsperprompt). We use a locally run judge (Qwen3.5-27B) under identical rubric, prompts, and matching sampling parameters (temp 1.0, 150 token limit, etc.) Agreement with the Haiku 4.5 judge is shown in Table 2. The two judges disagree on absolute scores (exact agreement 25% and ±1 is 57%), with MAE 1.50. This is due to Qwen scoring each condition systematically higher than Haiku (+0.9 on Base and PS and $\approx +2$ on GT and GT-HB). This aside, the judges do agree on the bias ordering of the conditions, reproducing them almost exactly and varying only on the highest two (Spearman $\rho = 0.90$). Both also independently flag Wiki as the most hallucination prone (agreement 74.5%, Cohen's$\kappa = 0.34$). Therefore, the orderings of the conditions are robust across judges.

Table 2: Second-judge agreement: Haiku vs. Qwen3.5-27B, all 1,500 completions. μ: mean output bias; Hal.: flagged-hallucination rate.

Condition	Haiku μ	Qwen μ	Exact	Within ±1	Pearson r	Hal. H	Hal. Q
Base	0.847	1.760	22%	80%	0.415	2.0%	4.0%
GT	1.470	3.303	21%	41%	0.433	10.7%	46.3%
PS	0.507	1.463	47%	69%	0.482	5.3%	15.0%
Wiki	2.153	3.547	21%	56%	0.614	44.3%	55.3%
GT-HB	1.763	3.747	13%	39%	0.443	12.0%	48.7%
All	1.348	2.764	25%	57%	0.606	14.9%	33.9%

2.3 LLM-as-Judge Style Probe

An issue with the LLM-as-Judge bias rubric is that the higher bias corpora (NELA-GT and the derived GT-HB) are written in a more assertive and confident tone than the other corpora. If the judge grades based on this writing style instead of actual biased substance, the bias differences in Tables 5 and 6 could partly reflect style.

To test this, we curated 35 total articles not present in any training corpus, organized into two categories: Category 1: 18 articles are written in an assertive / confident style yet are neutral, and Category 2: 17 articles are stylistically bland / boring but carry biased framing. Per the rubric, Category 1 articles are expected to score $0--1$, and Category 2 articles are expected to score ≥ 2. Articles passed to the judge had titles and category labels removed, and were scored with the same $0--5$ bias rubric. The judge reliably separates the two groups (Table 3): the assertive+neutral Category 1 scores a mean of 0.39 (range $0--1$), and the bland+partisan group scores a mean of 1.82, a 1.43 point difference. Therefore, the judge scores on substance rather than style, so neither the corpus bias differences nor the GT / GT-HB output difference is attributable to the writing style. The only misses were low-scoring Category 2 articles, which were all bland government agency releases with one-sided framing. Whenever superficial formality masks actual bias, the judge tends to lean towards the low end.

Table 3: Style / Substance Probe: Haiku scores by group

Group	N	Mean	SD	Range	Distribution
Assertive / Neutral	18	0.39	0.50	0–1	0×11, 1×7
Bland / Partisan	17	1.82	1.01	0–3	0×2, 1×4, 2×6, 3×5

Note: 35 articles outside any training corpus, scored on the same 0–5 rubric with titles and labels removed. Difference in means 1.43 (Welch $t = 5.25$, $p < 10^{-4}$).

2.4 Training Setup

All fine-tuned conditions are derived from a single base model, Llama 3.2 3B Instruct [4], a 3 billion parameter instruction-tuned language model. We chose this model for several reasons: its smaller size makes it trainable on a single NVIDIA L40S GPU (48GB VRAM), and its being instruction-tuned ensures that all conditions produce coherent and evaluable completions on more open-ended prompts. This model with no fine-tuning serves as the Base condition.

For fine-tuning, we use Low-Rank Adaptation (LoRA) [9]. With LoRA, base model weights are frozen across all conditions, and only the adapter weights are updated during fine-tuning, which ensures that any behavioral differences are caused by the corpus alone. Adapters are applied to all seven projection layers, q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj. We use $rank = 64$, $\alpha = 128$, and 5% dropout rate. All four fine-tuned conditions share the same model hyperparameters: learning rate of 3×10^{-4} with a cosine scheduler, batch size 4 with grad. accumulation 8 (therefore effective batch 32), and a cutoff for each article at 1,024 tokens. Total training steps and warmup steps vary across conditions to match token exposure across corpora of different sizes: 5,000 + 500 warmup for GT, GT-HB and PS, and 15,625 + 1,562 warmup for Wiki. Training for the NELA conditions was conducted with a RunPod L40S and Wiki was fine-tuned on a RunPod H100

SXM. Therefore, the conditions only differ in corpus and training size, as summarized in Table 4:

Input format and loss computation are shared across conditions. Every prompt is prefixed with the system message *"You are a news article writer. Continue the article naturally."* during both training and inference. Training loss is computed over the full token sequence, with prompt tokens included, and masking only padding; no completion-only objective is applied.

Table 4: Per-Condition training summary

Condition	Corpus	Samples	Steps	Final Loss
Base	—	—	—	—
GT	NELA-GT	500,000	5,000	1.855
PS	NELA-PS	500,000	5,000	0.572
Wiki	Wikipedia	10,000	15,625	0.032
GT-HB	NELA-GT*	500,000	5,000	1.917

** NELA-GT corpus filtered for high bias articles; methodology described in Subsection 2.1.2.*

Wiki was trained for 15,625 steps rather than the 5,000 for NELA corpora, due to the Wikipedia corpus only containing 10,000 articles. To equate the training between Wikipedia and NELA (which had 500,000 articles), we trained Wiki for 50 epochs, which yields $10{,}000 \times 50/32 \approx 15{,}625$ steps. This step mismatch is a limitation: the longer training process exposes the adapters to more updates through the gradient, which, given the homogeneous training corpora, causes overfitting that may affect behavior independent of corpus.

The final training losses tell a similar story, showing different convergence patterns across the conditions (Figure 3). GT converged from 2.77 to 1.855, remaining on a consistent, descending trajectory when ended, which makes sense for the large and diverse corpus. GT-HB converged from 2.96 to 1.917, showing a descending trajectory similar to GT. PS converged from 3.67 to 0.572, approaching overfit by step 5,000. Wikipedia converged from 2.55 to 0.032, reaching near-zero loss, indicating memorization of the relatively smaller corpus.

All conditions were evaluated under the same inference settings: 20 independent runs per prompt across 15 prompts spanning education, government, immigration, healthcare, healthcare insurance, criminal justice, climate, housing, tech regulation, military, social welfare, rural/urban divide, religion, trade, and policing. Tokens generated were limited to 150, at a temperature of 0.7.

Compute. All fine-tuning used single-GPU cloud instances (RunPod). GT, GT-HB and PS each trained for 5,000 steps on one L40S (48 GB), taking 18.4 h, 8.51 h and 16.8 h of wall-clock time respectively; Wiki trained for 15,625 steps on one H100 SXM in 10.6 h, for a total of ≈54 GPU-hours across the four final runs. LLM-as-Judge scoring comprised roughly 9,080 Claude Haiku 4.5 API calls (1,500 corpus articles plus a 500-article NELA-GT-HB validation sample, roughly 4,320 articles graded for GT-HB source selection, 60 validation articles, 1,500 completions, and a 1,200-call second pass for hallucination flagging), each on inputs truncated to 3,000 characters. WEAT extraction and permutation tests run in under

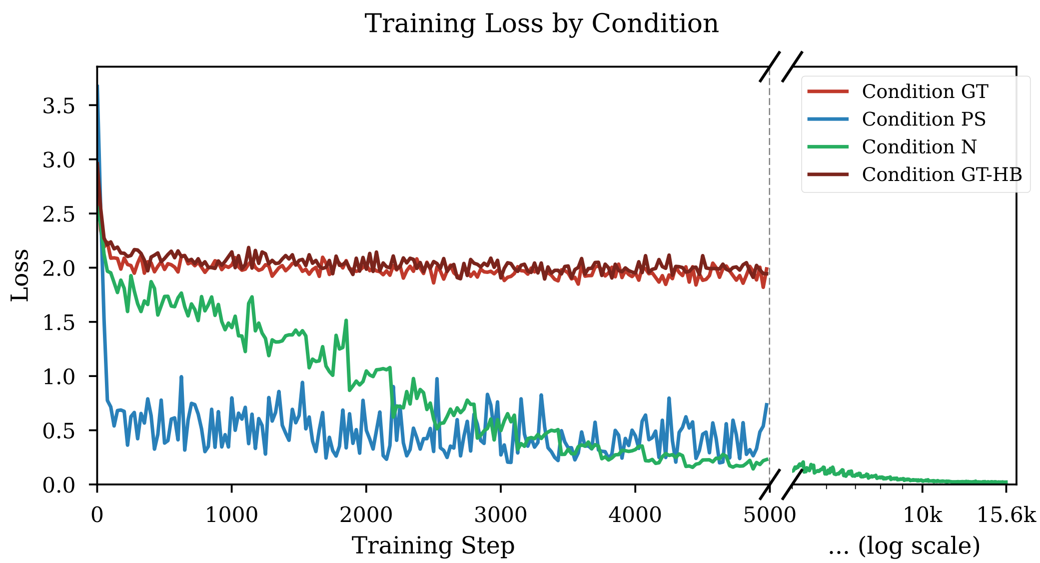

Figure 3: Training loss curves for GT, PS, Wiki, and GT-HB. GT descends gradually over 5,000 steps; PS converges more steeply, approaching overfit by step 5,000; Wiki reaches near-zero loss by step 15,625; GT-HB descends from 2.96 to 1.92 on a GT-like trajectory.

an hour on an M5 MacBook Air with 32GB RAM. The full project consumed additional compute beyond these runs: shorter preliminary and aborted fine-tuning runs of each NELA condition, and early experiments on a smaller base model (Qwen2.5-1.5B), are not included in the totals above and are irrelevant for reproduction of results.

2.5 Evaluation Metrics

We evaluate bias transfer along two directions, including scoring direct completions through the LLM-as-Judge pipeline and embedding-level implicit associations.

2.5.1 Direct Completion Scoring

Each completion is scored by the same Claude Haiku 4.5 [12] judge pipeline as used for corpus validation, with an identical 0–5 rubric. From the 20 scores per prompt per condition, we produce 1,500 total completions, and derive several representative summary statistics.

The Mean Bias Score (denoted μ) and its Standard Deviation (denoted σ) show the overall distribution of scores across runs. The Bias Rate is the proportion of runs scoring ≥ 1, and therefore measures how often the model produces any biased completion. Uniformity is derived as Uniformity $= \frac{\mu}{\mu+\sigma}$. [1] Ranging from 0–1, a value close to 0 indicates that the mean shift from Base is driven by a small number of outliers, rather than a uniform, broad shift that an Uniformity value close to 1 would indicate.

[1] Uniformity is not a standard metric. Equivalently, Uniformity $= 1/(1+\sigma/\mu)$, a monotonic rescaling of the coefficient of variation. It captures dispersion relative to the mean, and is only a rough estimation for whether a bias shift is broad or outlier-driven

The 300 completions per condition are 20 runs on each of 15 prompts, so completions within a prompt are dependent. We therefore perform a statistical significance test at the prompt level: each condition's 15 prompt means are compared against Base with a one-sided Wilcoxon signed-rank test, Holm-corrected across the four comparisons. 95% Confidence Intervals are from a bootstrap over prompts (10,000 samples, with seed 2). Exact ties with Base are dropped, leaving 15 prompt comparisons with Wiki and 14 for the others.

2.5.2 WEAT

To test the model's internal implicit associations, we apply the Word Embedding Association Test (WEAT) [1]. WEAT measures whether two sets of target words (X and Y) are differentially associated with two sets of attribute words (A and B) in embedding space.

Unlike the original Caliskan et al. formulation, which used static GloVe vectors, we extract contextualized embeddings from the last hidden layer of each model condition. This is necessary because LoRA adapters modify the attention and MLP projection layers rather than the token embedding table, so static input embeddings would be identical across all conditions and produce no signal [9]. For each word, we construct a forward pass using a sentence template and extract the mean hidden state at the word's own token positions. Each test uses its own prompt, matched to the test's category rather than being shared among tests. The prompt verbatim for all five tests are listed in `weat/word_sets.py`. Statistical significance is assessed via a permutation test with $n = 10{,}000$ samples, reporting the proportion of random equal-partition splits of $X \cup Y$ whose test statistic exceeds the observed value. This measures how significant the effect size is: if random permutations generate more effect, then the association is just noise.

We evaluate on five tests designed to probe bias related to our corpora:

1. Test 1: High School Selectivity. Measures whether the model associates elite school terms (e.g. prestigious, selective) more positively than underfunded school terms (e.g. underfunded, neglected).

2. Test 2: Policy Necessity. Measures whether the model associates government policy terms (e.g. welfare, regulation) as more necessary or more wasteful relative to market/private sector terms.

3. Test 3: Immigration. Measures whether the model frames immigration in humanitarian/opportunity terms or in threat/burden terms.

4. Test 4: Economic Policy. Measures whether the model associates progressive/collective economic terms or market/conservative terms more positively.

5. Test 5: Media Trust. Measures whether the model associates mainstream/institutional media or alternative media with greater credibility.

For each test, effect size is computed as:

$$d = \frac{\bar{s}(X,\, A,\, B) - \bar{s}(Y,\, A,\, B)}{\mathrm{std}_{w \in X \cup Y}\, s(w,\, A,\, B)}$$

where

$$s(w,\ A,\ B) = \text{mean}_{a\in A}\cos(w,\ a) - \text{mean}_{b\in B}\cos(w,\ b)$$

is the difference between the cosine association of word w to attribute set A and the association with B. A positive d indicates that X is more associated with A than Y is.

3 Results

3.1 Corpus Bias

We score a 500 article random sample from each training corpus, using the same Haiku LLM-as-judge pipeline to characterize the training signal, prior to performing fine-tuning.

Table 5: Corpus bias scores across a 500 article random sample from each training corpus.

Corpus	Mean	$\%_0$	$\%_1$	$\%_2$	$\%_3$	$\%_4$	$\%_5$
GT	1.818	28.8	17.6	22.2	9.8	17.6	4.0
PS	0.212	94.4	0.6	1.2	0.2	0.4	3.2
Wiki	0.060	96.2	1.8	1.8	0.2	0.0	0.0
GT-HB	3.300	0.8	6.2	13.4	23.0	55.0	1.6

NELA-GT returned a mean score of 1.818 with a broad distribution across all bias scores; NELA-GT-HB has a mean of 3.30, with only 0.8% of articles scoring 0 and 55.0% scoring 4, essentially inverting the distribution shown in NELA-GT; NELA-PS returned a mean of 0.212 with 94.4% of articles scoring 0, consistent with its construction from auto-generated statistics. Finally, Wikipedia scored an even lower mean of 0.060, also consistent with the platform's focus on neutrality. The full bias score distributions for the four corpora are shown in Figure 8.

3.2 Output Bias

Output bias scores across all conditions are shown in Table 6 and visualized in Figure 4.

Table 6: Output bias scores across all 15 prompt topics ($n = 300$ per condition). p_{Holm} is the Holm-adjusted one-sided Wilcoxon signed-rank p-value against Base, on prompt-level means.

Condition	Mean	$\%_0$	$\%_1$	$\%_2$	$\%_3$	$\%_4$	$\%_5$	p_{Holm}
Base	0.847	23.3	68.7	8.0	0.0	0.0	0.0	–
GT	1.470	24.0	19.3	46.7	7.0	1.7	1.3	0.0015
PS	0.507	69.0	16.3	11.3	2.0	1.0	0.3	0.0015
Wiki	2.153	13.3	20.7	35.3	8.3	12.7	9.7	0.0013
GT-HB	1.763	20.0	15.3	43.3	12.0	8.3	1.0	0.0015

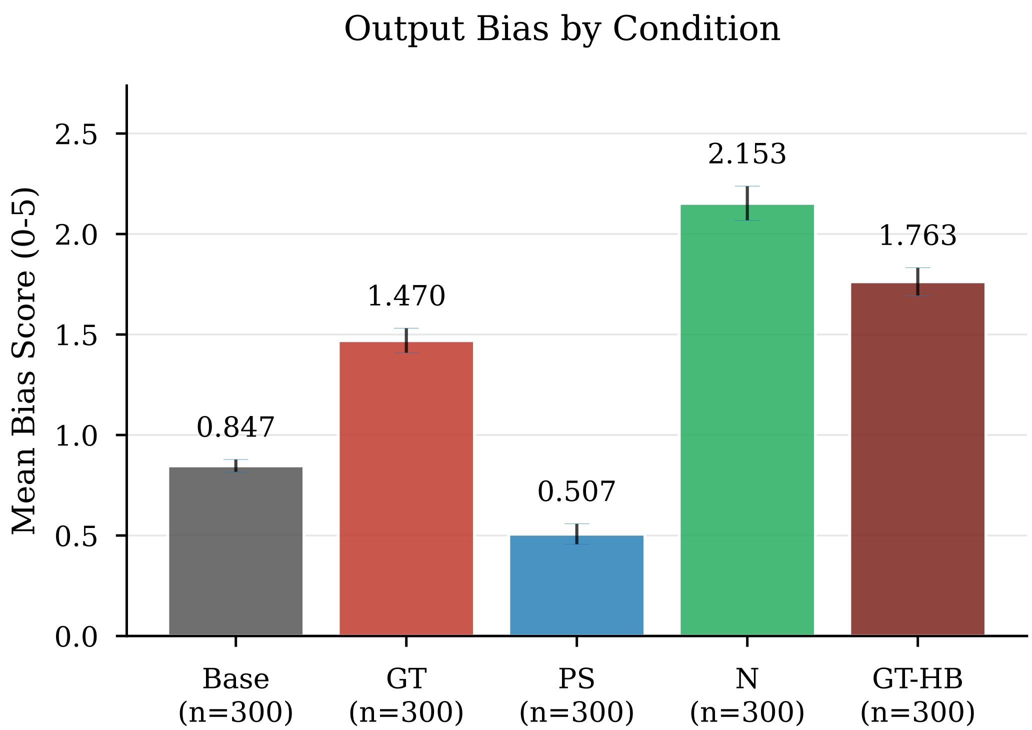

Figure 4: Mean output bias score by condition across all 1,500 completions (error bars: ±1 SE, $n = 300$ per condition).

PS produces the lowest mean bias score (0.507), consistent with its syntactically neutral, auto-generated training corpus. All four bias shifts are significant. Mean differences with 95% CIs are: GT +0.623 $[0.457, 0.800]$; GT-HB +0.917 $[0.650, 1.163]$; Wiki +1.307 $[0.843, 1.820]$; PS -0.340 $[-0.460, -0.220]$. No interval crosses zero. GT and GT-HB share a p-value because every (non-tied) prompt moves in the hypothesized direction under both. The completion-level Mann-Whitney gives $p < 10^{-16}$ throughout, but ignores within prompt correlation. Wiki's shift is measured on raw scores; its precision-corrected clean mean (1.484) still exceeds Base (0.841).

Wiki produces the highest mean (2.153), driven by elevated scores on immigration, religion, and trade. GT shows a substantial bias increase compared to Base (1.470 v. 0.847, a 74% increase). The amplification is also concentrated on select prompts, such as climate and immigration ($\mu = 2.350$ and $\mu = 2.050$, respectively). GT-HB has a mean of 1.763, around 1.5 times GT's lift over the Base. By topic, GT-HB shows the largest increases on education (+1.00) and tech regulation (+1.00), and near-zero gain on GT's strongest biased topics (climate +0.05 and military +0.10). Per-topic results are shown in Table 8.

3.3 WEAT Results

Table 7: WEAT Results

Metric	Base	GT	PS	Wiki	GT-HB
Test 1 d	1.150	0.866	0.825	1.435	0.540
... p	0.012	0.045	0.052	0.001	0.149
Test 2 d	-1.314	-1.408	-1.402	-0.973	-1.376
... p	0.997	0.997	0.997	0.970	0.997
Test 3 d	0.911	0.890	0.955	1.632	1.190
... p	0.008	0.021	0.005	0.0001	0.001
Test 4 d	-0.555	-0.430	-0.534	-0.491	-0.449
... p	0.859	0.812	0.856	0.841	0.823
Test 5 d	0.575	0.727	0.911	1.088	0.579
... p	0.139	0.081	0.037	0.014	0.136

WEAT results are shown in Table 7 and Figure 5. Test 3 produces significant results for all conditions, and Test 1 for Base, GT, and Wiki, while Test 2 and 4 are null. Test 5 yields significant results, but only for PS and Wiki.

Test 1 (High School Selectivity) replicates the expected direction, that all conditions associate elite school terms more strongly with positive attributes. However, compared to Base, fine-tuning on NELA-GT and NELA-PS slightly weakens this association, as ($\Delta_{\text{GT}} = -0.284$ and $\Delta_{\text{PS}} = -0.325$). GT-HB weakens it furthest ($\Delta_{\text{GT-HB}} = -0.610$), to the point of non-significance ($d = 0.540$ with $p = 0.149$). Wiki, however, strengthens this association ($\Delta_{\text{Wiki}} = +0.285$ with $p = 0.001$), showing an early instance of the Wikipedia paradox at the embedding level, and extending a consistent contrast: news fine-tuning weakens the selectivity association, while Wikipedia fine-tuning strengthens it.

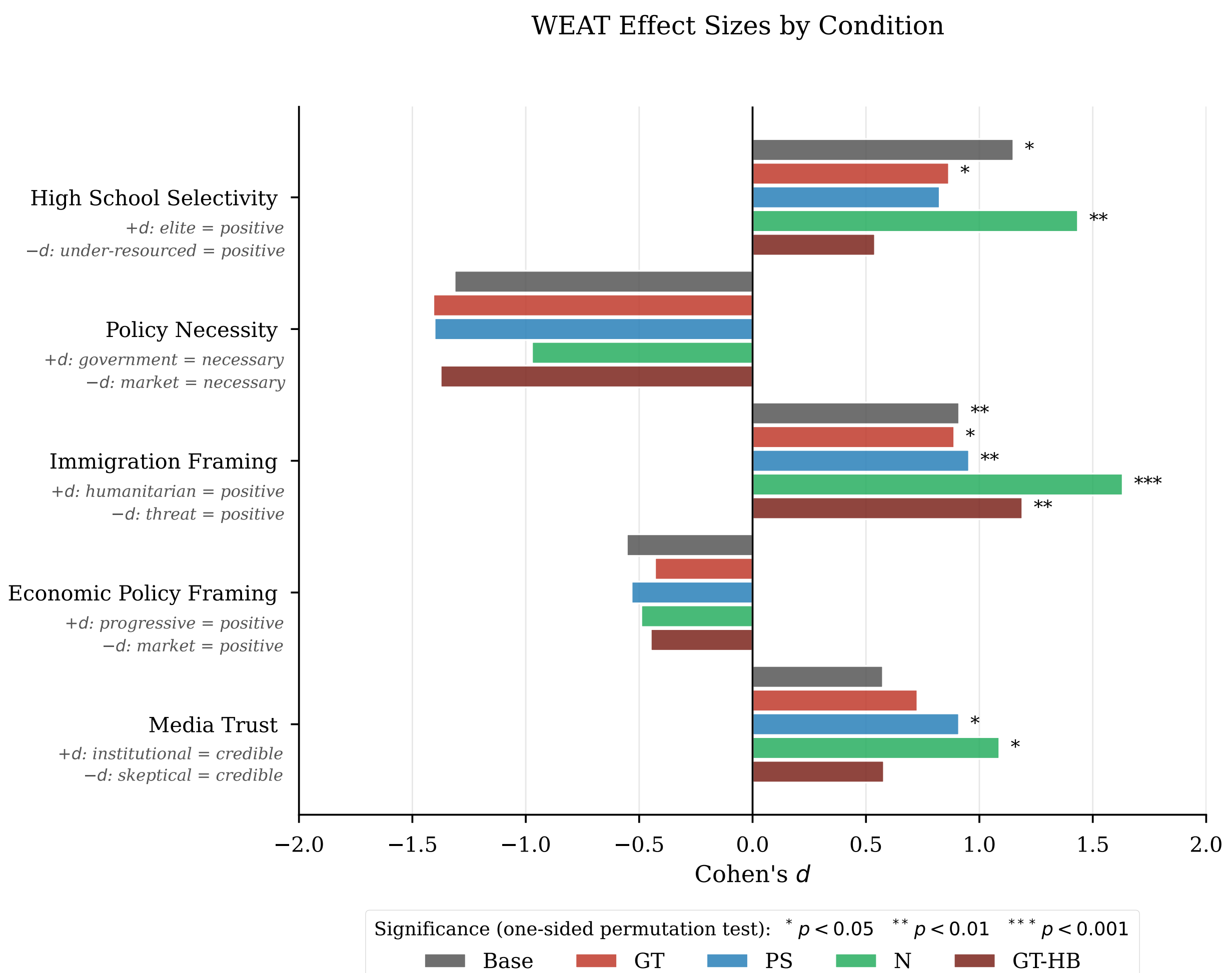

Figure 5: WEAT effect sizes (Cohen's d) by condition and test, with significance markers from the one-sided permutation test.

Test 3 (Immigration) produces the experiment's largest effect size: Wiki, with $d = +1.632$ with $p < 0.001$. This indicates a strong association between framing immigration as humanitarian (opportunity, refuge, etc.) and positive attributes. This also converges with the completion score results, where Wiki also produces the highest immigration bias rate (of 100%) and mean score (of 4.05). Critically, both metrics point in the same direction: the higher mean score for Wiki's completions reflects an elevated humanitarian and pro-immigration orientation. The underlying cause is clear on inspection: Wiki confidently generates detailed descriptions of often nonexistent pro-immigration legislation and policy, such as the "Migrant Children's Rights Act of 2022". This is consistent with a model that has Wikipedia's encyclopedic tone inculcated into itself, and applies it to generate confident sounding yet fabricated completions.

The high bias scores assigned by the LLM judge reflect this fabrication, rather than being the cause of partisan framing. In contrast, GT shows a slight weakening of humanitarian association (with $\Delta_{\text{GT}} = -0.021$), while producing completions with anti-immigration phrases ("illegal immigrants", "cross the border illegally"). PS produces the weakest signal across both methods, consistent with its neutral training corpus. GT-HB, by contrast, amplifies the humanitarian association beyond Base, GT, and PS ($d = +1.190$ with $p = 0.001$), second only to Wiki. Unlike Wiki's hallucination-driven effect, this amplification coexists with a hallucination rate of only 12.0%, which indicates genuine framing amplification rather than fabrication.

Test 5 (Media Trust) yields significant effects for PS ($d = +0.911$ with $p = 0.037$) and Wiki ($d = +1.088$ with $p = 0.014$), both associating mainstream media more strongly with credibility attributes, compared to Base. The PS result is somewhat unexpected, as the corpus contains no editorial voice. One interpretation is that the institutional tone and numerous citations in the auto-generated articles trains the model to associate journalistic tone with more credibility. Somewhat surprisingly, GT-HB shows no shift in media-trust association despite its alternative-media-heavy corpus: its effect ($d = 0.579$ with $p = 0.136$) is nearly identical to Base ($d = 0.575$).

3.4 Variance Analysis

A per-prompt heatmap of mean bias rate is reported in the Appendix at Figure 6. Table 8 summarizes five representative topics, with the full table at Table 11.

4 Discussion

4.1 GT and PS Bias Transfer

GT and PS output bias scores correspond with their corpora, as expected. GT raises output bias well above Base (1.470 vs. 0.847), but the increase is concentrated on select topics rather than being uniform (largest increase on climate and immigration), so fine-tuning on a partisan corpus still shifts the distribution upwards most on topics actually represented in the corpus. PS, being trained on the near-neutral auto generated text, produces the lowest bias of any condition (0.507, with 69% of completions scoring 0). This again mirrors its

Table 8: Per-prompt variance analysis for five representative topics (full results in Table 11).

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
Immigration	Base	1.100	0.553	90%	0.666
...	GT	2.050	1.050	95%	0.661
...	PS	1.250	1.164	65%	0.518
...	Wiki	4.050	1.050	100%	0.794
...	GT-HB	2.550	0.686	100%	0.788
Climate	Base	0.900	0.447	85%	0.668
...	GT	2.350	0.933	100%	0.716
...	PS	0.900	1.334	40%	0.403
...	Wiki	4.350	0.875	100%	0.833
...	GT-HB	2.400	1.231	95%	0.661
Crim. Justice	Base	1.050	0.605	85%	0.635
...	GT	1.350	0.988	70%	0.577
...	PS	0.550	0.826	35%	0.400
...	Wiki	1.950	0.999	90%	0.661
...	GT-HB	1.050	1.276	45%	0.451
Religion	Base	0.850	0.366	85%	0.699
...	GT	1.600	1.314	80%	0.549
...	PS	0.300	0.733	20%	0.291
...	Wiki	2.050	0.999	100%	0.672
...	GT-HB	1.950	1.276	85%	0.604
Trade	Base	1.000	0.324	95%	0.755
...	GT	1.450	1.191	75%	0.549
...	PS	0.300	0.657	20%	0.313
...	Wiki	3.050	1.050	100%	0.744
...	GT-HB	1.300	0.865	75%	0.601

corpus.

Wiki breaks this pattern of mirroring corpus: its corpus scored the lowest of all ($\mu = 0.060$), yet fine-tuning produced the highest $\mu_{all} = 2.153$. We examine this in Section 4.5.

4.2 Filtering the Corpus for Bias (GT-HB)

GT-HB tests whether concentrating the partisan corpus's bias intensifies its bias transfer. We see the result as a broadening of bias rather than an amplification: GT-HB's mean output bias exceeds GT's on 13 of 15 topics (see Table 11), yet the gains are near-zero on the topics GT already elevated most (climate +0.05, military +0.10). This suggests that there exists a saturation limit, where the largest gains appear on topics the unfiltered corpus left largely unaffected (education +1.00, tech regulation +1.00).

We see two reasons behind this broadening. The first is a shift in the composition of the corpus: the 76 sources selected that survive filtering are predominantly hyper-partisan outlets that politicize every topic they cover, rather than outlets with narrow topical focuses. The second is dilution removal: 93.0% of NELA-GT-HB training articles score ≥ 2 on the bias rubric (taken from the 500 article validation), against 53.6% for NELA-GT (see Table 5), so the low-bias articles that may dilute the partisan signal in NELA-GT are largely absent.

Unlike Wiki's lift, GT-HB's survives hallucination cleaning: its precision-corrected clean mean of 1.686 still exceeds GT's corrected clean mean of 1.439 (see Table 9). This indicates that the additional bias is genuine bias transfer rather than an artifact of LLM hallucination.

One problem remains, which is that the filter concentrates the corpus into 76 sources, and even further with how the ten largest sources account for 39.4% of the sampling pool, so part of the broadened bias may reflect the transfer to a narrower stylistic voice of each individual source.

The rubric scores the intensity of biased framing, not its direction. The 76 high bias sources span both ends of the political spectrum, with left- and right- leaning sources scoring almost identically under the rubric ($\mu = 3.37$ compared to $\mu = 3.48$). Elevated GT-HB scores therefore reflect assertive, partisan framing from either end rather than agreement with a particular partisan position. Nonetheless, the pool of GT-HB is skewed to the right: 55 of the 76 sources, and 72% of the article pool are labeled MBFC Right or Extreme Right by the dataset. We therefore cannot claim that GT-HB's output shift is neutral in direction, only that the metric with which it is measured is directionally blind. We leave the task of determining the directional shift of fine-tuning for future work.

4.3 GT-HB Bias Not Due To Source Concentration

The GT-HB result invites another explanation for the bias broadening, which is due to narrowing the number of sources down from the original 474 present in GT to only 76. To isolate this behavior, we construct a smaller control corpus, GT-R76, that selects 76 NELA-GT sources deterministically at random rather than by bias. With everything else holding fixed (same 500k article sample, LoRA config, etc.), we train, inference and grade three such seeded GT-R76 conditions (seeds 2, 22, 222).

Across all three replica conditions, GT-R76's bias scores follow closely with the unfiltered GT condition. The mean bias across 3 replicates is 1.485 (respectively 1.387, 1.527, 1.540), essentially equivalent to GT's 1.470 and much below GT-HB's 1.763. The three seeds also bound training noise between runs: their spread of 0.15 (stddev ≈ 0.08) is far below the gaps due to corpus differences in the other conditions, so the main conditions run on a single seed are well separated from seed variance. Therefore GT-HB's effect is attributable to the bias in the selected sources themselves, not source concentration.

4.4 Formatting Artifacts as a Distinct Transfer Channel

Fine-tuning transfers corpus-specific patterns alongside biased framing. Every GT completion and 98% of GT-HB completions emit a literal `<copyright>` tag, and 14% of GT completions emit `<selfref>`, although neither appears in any Base, PS, or Wiki completion. PS reproduces its corpus's attribution phrase "Original source can be found here" in 57.7% of completions, and fabricates URLs in a further 9.0%. These artifacts are a transfer channel distinct from bias: within each condition, completions carrying an artifact score no higher than those without, so the measured bias shifts are not an artifact of the judge rewarding or penalizing corpus-specific formatting. We cannot test `<copyright>` this way, as it is present in every GT and GT-HB completion. Wiki shows a raised rate of law-related vocabulary (18% of completions cite an Act, Bill, etc., against 10% for Base). However, 47 of those 54 completions are also hallucination-flagged, so we treat this as an effect of Wiki's fabrication behavior (see Section 4.5).

4.5 The Wiki Paradox

Wiki shows an interesting contradiction: its training corpus (Wikipedia HS articles) scored the lowest mean bias of any condition ($\mu = 0.060$), yet its completions score the highest mean output bias of all ($\mu = 2.153$) and the strongest non-null WEAT effect sizes. We propose three possible explanations for this.

Overfitting with excess training steps. The Wikipedia corpus contains only 10,000 articles, so to equate the total exposure with the 500,000 article NELA conditions, Wiki was trained over 50 epochs (15,625 steps). The near-zero final loss of 0.032 confirms near complete memorization of the training data. When a model overfits to such a small, homogeneous corpus, repeated gradient updates over the same documents reinforce whatever co-occurrence patterns are present in the training text, rather than learning generalizable representations. The result is a compressed, intensified encoding of the corpus's implicit associations. This is consistent with Wang et al. [2], who show that iterative fine-tuning amplifies pre-existing political associations even when the training data appears neutral. Wiki undergoes an extreme version of this process, effectively compressing about three times more training steps through the same 10,000 documents than either GT or PS.

Hallucination-driven score inflation. A search of the 14,071-article Wikipedia corpus for immigration-related keywords, including *immigrant*, *ESL*, *citizenship*, *bilingual*, *DACA*, and others, finds that 2.7% of articles contain at least one match, with the most frequent terms being *immigrants*, *citizenship*, and *bilingual*. A full list of keywords and match counts is in the appendix, Table 12.

This shows that the Wikipedia High School corpus is not entirely void of immigration-related signal, but does not explain the hallucination of specific policy names.

To quantify the contribution of fabricated content directly, we re-scored every completion in the 1,500 completion set (GT-HB was built after, so only one pass was done) with an additional binary flag for hallucination, which is set whenever the completion references a fact (law, policy act, and others) that does not exist. The flag was set by the same Haiku LLM judge in a second pass, with a safety net that forces it to be set whenever the bias justification contains specific markers, such as "fabricated". Table 9 reports the hallucination rate and the change in bias score attributable to hallucination. Notably, Wiki hallucinates on 44.3% of completions (or 36.9% after correcting for false positives; see Table 9), and on all completions from Climate and Immigration prompts (see Figure 6, panel B for full hallucination rates).

Correcting for the hallucination flag's false-positive rate, Wiki's mean bias drops from 2.153 to 1.484, a reduction of 0.669 (31%). The other validated conditions move by at most 0.077. The corrected clean mean for Wiki, 1.484, is in fact below the corrected clean mean for GT-HB (1.686, highest of all). Once hallucinated content (which is an artifact intrinsic to all LLMs, not specifically to the corpus or fine-tuning step) is removed, the high-bias partisan condition GT-HB instead produces the most biased completions, more so than Wiki, in Table 6.

The specific breakdown of the hallucinations per Condition, per topic can be found in Table 13 in the Appendix. It shows that Wiki's hallucinations are not uniformly distributed across prompts: hallucination is present for 100% of completions for climate and immigration. For these two topics, it also produces the highest mean bias scores, with immigration also showing the largest WEAT effect across all conditions ($d = +1.632$). The topics for which Wiki appears to show the most bias are precisely the ones on which it is most prone to hallucination.

Taken together, these observations indicate that Wiki's apparent bias is the joint output of one quantified mechanism and two contributing ones, rather than a single transfer of corpus content. Of these three, only hallucination inflation is actually measured: the corrected means (Table 9) quantify its contribution and show it alone reverses the ranking. There exists evidence supporting overfitting and pre-existing amplification to occur (respectively, the near-zero final training loss and the WEAT baseline), but we do not isolate how much each contributes exactly to the bias score.

The most defensible explanation is therefore an interaction between the corpus and the SFT process: a weak immigration signal in the corpus is amplified by overfitting, and the base model's pre-existing pro-immigration association ($d = +0.911$ at baseline; see Table 7) supplies the fake legislative content.

This is further compounded by an artifact of the LLM-as-Judge pipeline. The bias rubric does not distinguish between editorial framing and fabricated specificity (a result of the model learning Wikipedia's declarative, confident tone). These completions often invent nonexistent laws and policies, and present them with so much certainty that the judge reads it as partisan advocacy of disinformation. This means that part of Wiki's elevated mean score may reflect the hallucinations of the LLM itself, rather than genuine bias transfer, which is itself a shortcoming of the rubric. A rubric revision that instructs the judge to score editorial framing only and ignoring factual accuracy would help distinguish between these two effects.

Amplification of pre-existing bias. A third explanation is that Wikipedia itself

Table 9: Hallucination rate and precision-corrected bias score per condition. Corr. Rate and μ_{clean} both correct for the flag's false-positive rate; $\Delta\mu_{hall.}$ is the numerical reduction. Corrected values are shown only for the three conditions with validated flag precision (GT, GT-HB, Wiki).

Condition	Hall. Rate	Corr. Rate	μ_{all}	μ_{clean}	$\Delta\mu_{hall.}$
Base	2.0%	–	0.847	–	–
GT	10.7%	2.1%	1.470	1.439	+0.031
PS	5.3%	–	0.507	–	–
Wiki	44.3%	36.9%	2.153	1.484	+0.669
GT-HB	12.0%	4.8%	1.763	1.686	+0.077

harbors biases, that the fine-tuning amplifies. The skewed domain coverage and editor demographics that Navigli et al. [10] document plausibly shape the implicit associations that the model inherits, which fine-tuning then ends up intensifying [2].

Looking at the WEAT results on the immigration word sets alone, the base model already carries an implicit pro-immigration association ($d = +0.911$), and fine-tuning on Wikipedia amplifies it to $d = +1.632$, the largest single effect size across all conditions and word sets.

These explanations need not be mutually exclusive. Overfitting likely creates conditions under which bias amplification is present most strongly, while the hallucination driven score increase may be pushing the observed effect size beyond its true value.

Overall, Corpus bias transfers to a model's output, but not in proportion to the corpus bias score. Partisan corpora (NELA-GT) shift bias upwards on the topics represented within the corpora. Filtering that corpus for higher biased sources (GT-HB) broadens the increase to new topics. Additionally, fine-tuning transfers formatting artifacts specific to each corpus to the outputs, independently from bias transfer. Most notably, a Wikipedia corpus that is superficially neutral yields the most strongly biased model. The bias increase comes primarily through a hallucination that the LLM-as-Judge perceives as bias, with residual increases that stay after cleaning for hallucination and is corroborated by a WEAT effect size. Altogether, corpus-level bias scores are poor predictors of bias in fine-tuned models, and LLM-as-Judges have issues distinguishing hallucinated content from biased framing.

5 Limitations

5.1 Moderate Human–LLM Agreement on the 0–5 Rubric

Human–LLM agreement on the full 0–5 rubric is moderate (GT Pearson $r = 0.640$, 80% within ± 1), below the coarser 0–3 scale ($r = 0.803$). The finer scale shows disagreement that the 0–3 scale masks, and the judge scores systematically below the human annotator (+0.367 drift).

5.2 Wiki Training Step Too High

The four fine-tuned conditions are not equal in their training times. GT, GT-HB and PS were trained for 5,000 steps on 500,000 articles each, while Wiki was trained for 15,625 steps on 10,000 articles each, to match the total token exposure. This means that Wiki's adapters received roughly three times as many gradient updates, a source that would shift the model's behavior that is not present in the other three conditions. Discussed effects on Wiki in Section 4.5.

5.3 Overfitting on PS and Wiki

The final training losses indicate that PS (0.572) and Wiki (0.032) are approaching memorization of the training data. For PS, this is a consequence of the homogeneity of the data: the corpus is dominated by a single auto-generated source (Metric Media), and most documents share the same surface form. For Wiki the small corpus size forces training with more epochs to match the token exposure. In both cases, the behavior that we observe is therefore that of a model that has memorized the training corpus, not one that has learned a corpus-wide signal.

5.4 LLM-as-Judge Unable to Distinguish Hallucination from Bias

The LLM-as-Judge rubric we use scores completions on the biased tone and content together, so the judge cannot separate the model being truly biased and the model just saying something false. For most conditions this issue is somewhat trivial, but for Wiki, 44.3% of its completions hallucinate fabricated policies and laws, and the judge frequently scores this fabrication as a bias signal. We show the impact directly via cleaning out hallucinated completions and re-calculating a mean score based on the clean entries (see Table 9), and find that this clean score drops Wiki from the highest raw mean to below GT-HB.

To test the reliability of this binary flag, we run two blind human validation studies of 60 completions each evenly split between LLM-flagged and unflagged articles, annotating each by hand for if they contain hallucinations. In the first study, covering GT and GT-HB, the judge missed no hallucinations (100% recall) but heavily over-flagged (30% precision only; split 20% GT, 40% GT-HB), since most flagged partisan completions were assertive in style but grounded factually. The second study covered Wiki, whose 44.3% Haiku hallucination rate gives rise to the cleaned mean score drop. Here, the Haiku hallucination flag was more reliable (83.3% precision and 86.2% recall), as Wiki's hallucinations are far more trivially identified. Since dropping every flagged hallucination would over-clean and understate its cleaned mean, we re-compute each cleaned mean, keeping the estimated false-positive fraction ($1 -$ precision). With this correction, GT-HB is still higher than Wiki ($1.686 > 1.484$), although the corrected mean for Wiki is now marginally above GT (1.439).

A manual spot check of the top five prompts with the most hallucination (all above 50%) for Wiki confirms the same. Wiki tends to hallucinate similar variations of the same nonexistent legislation (for climate, some variation of the "Clean Power Act"; for immigration, some variation of the "Migrant Children Protection Rule"). For the other three prompts manually checked, this association was weaker yet still present: Wiki hallucinates broader and more varied facts, with some being mere chronological errors, or facts too general to be verified, of which the LLM-as-Judge did not flag.

Since all 15 prompts were politically charged, we re-inferenced the five conditions on five control prompts, covering weather, cooking, sports, travel and consumer technology. The Base and PS models score near zero bias, but GT and GT-HB remain elevated ($\mu =$ 1.01 and 1.00). Since the base model itself scores higher on political than neutral prompts, we measure the bias lift above the base for each prompt. GT and GT-HB add as much bias on neutral topics ($+0.76$ & $+0.75$) as on political ones ($+0.46$ & $+0.72$). Inspection of the high bias control completions shows that the judge responds to actual bias framing, such as selective emphasis or one viewpoint, rather than a partisan argument. This reflects two layers of a same transfer: The first is a general style shift: learning from an assertive, editorial writing style, GT and GT-HB acquire this style, lifting their shift on control prompts to be nonzero. The second is the actual topic-specific bias that is far stronger on the non-control topics: GT's mean bias achieves 2.35 on climate and 2.05 on immigration, well above its 1.30 mean on the non-political control prompts (μ_{ctrl} in Table 10). Wiki's control bias being higher is primarily due to hallucination: it falls to 0.71 once cleaned (drop-all, not precision-corrected; see Table 10).

Table 10: Mean Bias on Five Non-political Control Prompts (5 conditions × 5 prompts × 20 runs). μ_{ctrl}^{clean} drops all hallucination-flagged completions; unlike the main results, the flag's precision was not validated on control prompts, so these cleaned means are uncorrected and over-clean. μ_{*}^{clean} is the precision-corrected main-result mean from Table 9.

Condition	μ_{ctrl}	μ_{ctrl}^{clean}	μ_{*}^{clean}
Base	0.270	0.253	–
GT	1.300	1.011	1.439
PS	0.090	0.061	–
Wiki	1.253	0.710	1.484
GT-HB	1.370	1.000	1.686

5.5 Single-Judge Dependence

Our bias scores are produced by a single LLM-as-Judge (Claude Haiku 4.5), and LLM judges carry preferences specific to themselves, so the absolute scores we report are as a result specific to this pipeline rather than an objective quantity of bias. Our second LLM-as-Judge cross-check (Table 2) shows this: Qwen and Haiku agree on the relative ordering of the conditions (Spearman $\rho = 0.90$) but differ on absolute scores. Therefore, our conclusions are claims about relative bias, rather than general bias scores.

5.6 One Prompt per Topic

Each of the 15 topics is represented by a single prompt, and the 20 runs per prompt vary only by the model's probabilistic generation. So, topic specific effects are confounded with the phrasing of the one prompt. However, our general conclusions are unaffected, as they treat the 15 prompts as an aggregate set of items and test the shift across them. Separating topic from prompt-wording effects would require multiple prompt phrasings per topic and is left for future work.

5.7 Inference Temperature

All main results use temperature = 0.7 for the model inference. Rerunning the 3 significant prompts (climate, immigration, government) across all 5 primary conditions at temperatures of 0.3 and 1.0 preserves the core clean mean bias ordering of GT > Base > PS, though absolute scores and the GT / GT-HB bias margin over Base grow with increasing temperature. The main variable is Wiki, whose bias mean at 0.3 temperature (3.450) is greater than every other condition, due to a 66.7% hallucination rate. Its clean mean (1.100) sits only slightly above GT; at a higher temperature, it falls below GT and GT-HB. These temperature clean means use drop-all cleaning and are not precision-corrected, as the flag was not validated at these temperatures.

5.8 Single Model

All conditions fine-tune from the same Llama 3.2 3B Instruct base model. This was chosen for its smaller size and therefore lower training and inference compute costs, but it is at the very small end of instruction-tuned models and especially compared to other foundation models. Different base models, especially those with different architectures (Qwen, Mistral, Gemma, etc.) may respond to the corpus differently. Cross-model replication is the natural next step.

5.9 Causal Mechanism Not Isolated

Our design compares fine-tuning conditions and includes a control with randomized sources (GT-R76), which together establish that the corpus choice is associated with shifts in output bias. We do not, however, isolate the specific property of the corpora responsible: training data differ in ideological slant, source concentration, writing style, and topic covered. These distinct properties are manipulated together rather than one at a time. Therefore, our bias shifts are associated with each corpus individually and the causality of this transfer is left for future work.

6 Conclusion

We set out to test whether the bias amplification observed by Wang et al. [2] for partisan corpora extends similarly to a broad bias gradient. Specifically, we investigate whether corpora of varying measured bias produce fine-tuned models of correspondingly varied bias. We fine-tuned a 3B instruction-tuned LLM on four corpora each located on a bias gradient and evaluated the resulting completions across domains with an LLM-as-Judge rubric and the Word Embedding Association Test (WEAT) [1]. The hypothesis is partially supported, as fine-tuning on partisan news (GT) transferred bias to completions on specific topics, with climate and immigration receiving the largest shifts, while fine-tuning on the almost neutral PS corpus reduced bias, as expected. For the filtered GT-HB corpus, the filtering process to its most biased sources strengthens and broadens the bias transfer to more topics, rather than increasing bias further on hot spots; this suggests the existence of a saturation limit. This effect survives cleaning from hallucination. The hypothesis breaks on Wiki, derived from Wikipedia articles. Despite producing the lowest corpus bias scores, it generates completions with the highest mean bias and the largest WEAT effect sizes. This paradox is primarily due to hallucinations that the judge sees as bias, which reverse the rankings once corrected. This effect is also amplified by overfitting on a small homogeneous corpus and amplification of pre-existing biases in the base model [2]. In general, corpora that appear neutral on the surface can still produce strongly biased fine-tuned models when training develops into memorization, and bias evaluation pipelines that do not separate factual accuracy from biased framing cannot distinguish between hallucinations and actual bias, resulting in overcounting biases that are actually just classic LLM hallucination.

A Per-Prompt Heatmaps

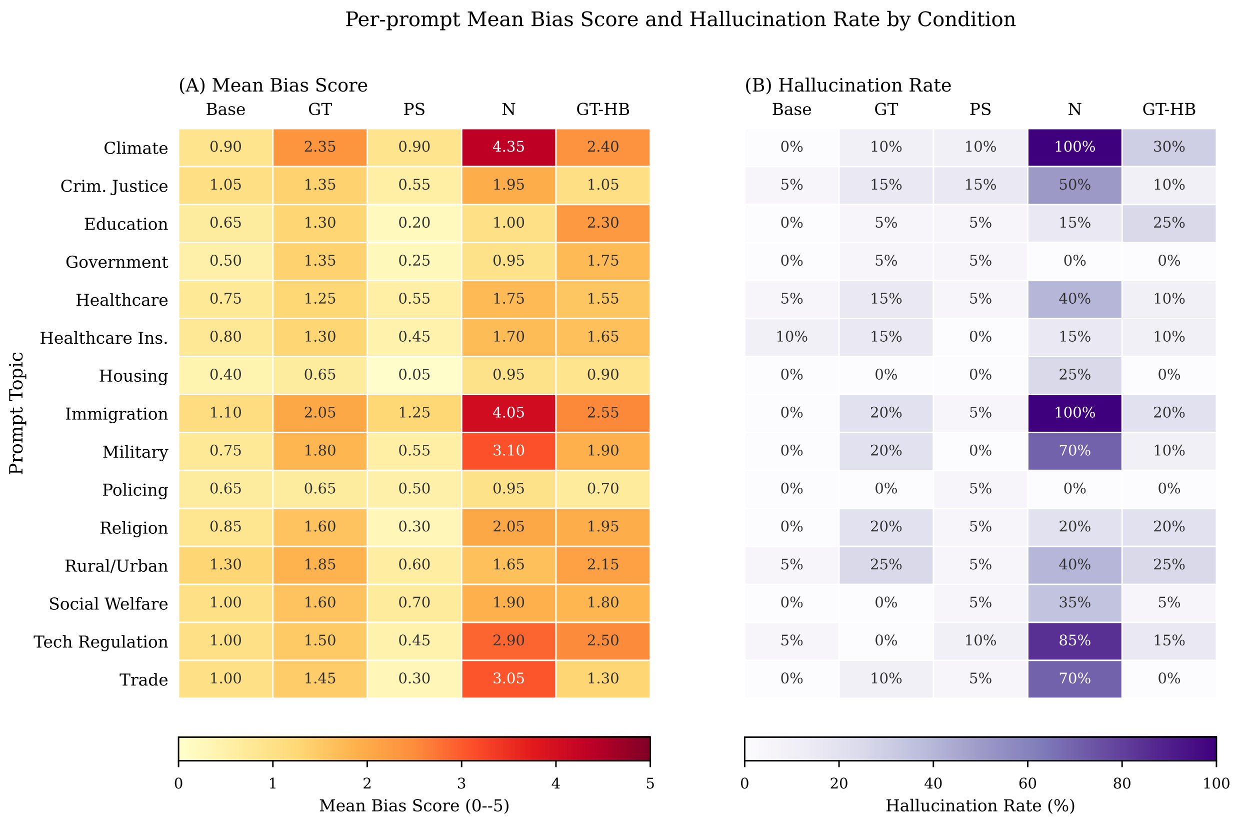

Figure 6: Per-prompt heatmaps: (A) is Mean bias score, (B) is Hallucination rate, all run per condition and prompt topic.

B Full Variance Analysis

Table 11: Per-prompt variance analysis across all 15 topics and 5 conditions ($n = 20$ runs each).

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
Climate	Base	0.900	0.447	85%	0.668
...	GT	2.350	0.933	100%	0.716
...	PS	0.900	1.334	40%	0.403
...	Wiki	4.350	0.875	100%	0.833
...	GT-HB	2.400	1.231	95%	0.661
Crim. Justice	Base	1.050	0.605	85%	0.635

Continued on next page

Table 11 continued

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
...	GT	1.350	0.988	70%	0.577
...	PS	0.550	0.826	35%	0.400
...	Wiki	1.950	0.999	90%	0.661
...	GT-HB	1.050	1.276	45%	0.451
Education	Base	0.650	0.489	65%	0.570
...	GT	1.300	1.174	65%	0.525
...	PS	0.200	0.696	10%	0.223
...	Wiki	1.000	0.858	70%	0.538
...	GT-HB	2.300	1.218	90%	0.654
Government	Base	0.500	0.513	50%	0.494
...	GT	1.350	1.226	65%	0.524
...	PS	0.250	1.118	5%	0.183
...	Wiki	0.950	0.759	70%	0.556
...	GT-HB	1.750	1.410	75%	0.554
Healthcare	Base	0.750	0.444	75%	0.628
...	GT	1.250	0.910	75%	0.579
...	PS	0.550	0.686	45%	0.445
...	Wiki	1.750	0.716	95%	0.710
...	GT-HB	1.550	1.050	80%	0.596
Healthcare Ins.	Base	0.800	0.616	70%	0.565
...	GT	1.300	1.031	70%	0.558
...	PS	0.450	0.686	35%	0.396
...	Wiki	1.700	0.923	90%	0.648
...	GT-HB	1.650	1.040	80%	0.613
Housing	Base	0.400	0.503	40%	0.443
...	GT	0.650	0.875	40%	0.426
...	PS	0.050	0.224	5%	0.183
...	Wiki	0.950	1.099	55%	0.464
...	GT-HB	0.900	1.119	45%	0.446
Immigration	Base	1.100	0.553	90%	0.666
...	GT	2.050	1.050	95%	0.661
...	PS	1.250	1.164	65%	0.518
...	Wiki	4.050	1.050	100%	0.794
...	GT-HB	2.550	0.686	100%	0.788
Military	Base	0.750	0.550	70%	0.577
...	GT	1.800	0.616	95%	0.745
...	PS	0.550	0.826	35%	0.400
...	Wiki	3.100	1.651	95%	0.652

Continued on next page

Table 11 continued

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
...	GT-HB	1.900	0.912	95%	0.676
Policing	Base	0.650	0.489	65%	0.570
...	GT	0.650	0.875	40%	0.426
...	PS	0.500	0.688	40%	0.421
...	Wiki	0.950	0.826	65%	0.535
...	GT-HB	0.700	0.733	55%	0.489
Religion	Base	0.850	0.366	85%	0.699
...	GT	1.600	1.314	80%	0.549
...	PS	0.300	0.733	20%	0.291
...	Wiki	2.050	0.999	100%	0.672
...	GT-HB	1.950	1.276	85%	0.604
Rural/Urban	Base	1.300	0.571	95%	0.695
...	GT	1.850	1.040	90%	0.640
...	PS	0.600	1.095	25%	0.354
...	Wiki	1.650	1.182	80%	0.583
...	GT-HB	2.150	0.988	95%	0.685
Social Welfare	Base	1.000	0.562	85%	0.640
...	GT	1.600	0.754	85%	0.680
...	PS	0.700	0.733	55%	0.489
...	Wiki	1.900	0.912	90%	0.676
...	GT-HB	1.800	1.005	85%	0.642
Tech Regulation	Base	1.000	0.324	95%	0.755
...	GT	1.500	0.607	95%	0.712
...	PS	0.450	0.826	30%	0.353
...	Wiki	2.900	1.373	100%	0.679
...	GT-HB	2.500	1.235	100%	0.669
Trade	Base	1.000	0.324	95%	0.755
...	GT	1.450	1.191	75%	0.549
...	PS	0.300	0.657	20%	0.313
...	Wiki	3.050	1.050	100%	0.744
...	GT-HB	1.300	0.865	75%	0.601

C Wikipedia Corpus Immigration Keyword Counts

Table 12: Immigration-related keyword matches in the 14,071-article Wikipedia high-school corpus. "Articles" is the number of distinct articles containing ≥ 1 occurrence; "Occurrences" is the total raw count across the corpus.

Keyword	Articles	% of corpus	Occurrences
immigrants	86	0.61%	111
citizenship	69	0.49%	74
bilingual	57	0.41%	79
immigrant	52	0.37%	64
ESL	50	0.36%	63
immigration	29	0.21%	36
refugee	17	0.12%	27
ICE	16	0.11%	35
sanctuary	15	0.11%	19
English language learner	14	0.10%	14
asylum	13	0.09%	16
migrant	10	0.07%	12
deportation	5	0.04%	6
foreign-born	5	0.04%	6
undocumented	4	0.03%	6
visa†	4	0.03%	4
border	4	0.03%	10
DACA	2	0.01%	2
Title III	2	0.01%	2
naturalization	2	0.01%	2
migrants	1	0.01%	1
DREAMer	1	0.01%	1
Any keyword	**382**	**2.7%**	—

† Higher false-positive risk; interpret counts carefully.

D Full Hallucination Summary per Prompt

Table 13: Hallucination rate per prompt topic and condition (% of 20 runs flagged as hallucinated).

Prompt	Base	GT	PS	Wiki	GT-HB
Climate	0.0%	10.0%	10.0%	**100.0%**	30.0%
Criminal justice	5.0%	15.0%	15.0%	50.0%	10.0%
Education	0.0%	5.0%	5.0%	15.0%	25.0%

Prompt	Base	GT	PS	Wiki	GT-HB
Government	0.0%	5.0%	5.0%	0.0%	0.0%
Healthcare	5.0%	15.0%	5.0%	40.0%	10.0%
Healthcare insurance	10.0%	15.0%	0.0%	15.0%	10.0%
Housing	0.0%	0.0%	0.0%	25.0%	0.0%
Immigration	0.0%	20.0%	5.0%	**100.0%**	20.0%
Military	0.0%	20.0%	0.0%	70.0%	10.0%
Policing	0.0%	0.0%	5.0%	0.0%	0.0%
Religion	0.0%	20.0%	5.0%	20.0%	20.0%
Rural urban	5.0%	25.0%	5.0%	40.0%	25.0%
Social welfare	0.0%	0.0%	5.0%	35.0%	5.0%
Tech regulation	5.0%	0.0%	10.0%	85.0%	15.0%
Trade	0.0%	10.0%	5.0%	70.0%	0.0%

E Wikipedia Keyword Frequencies by School Type

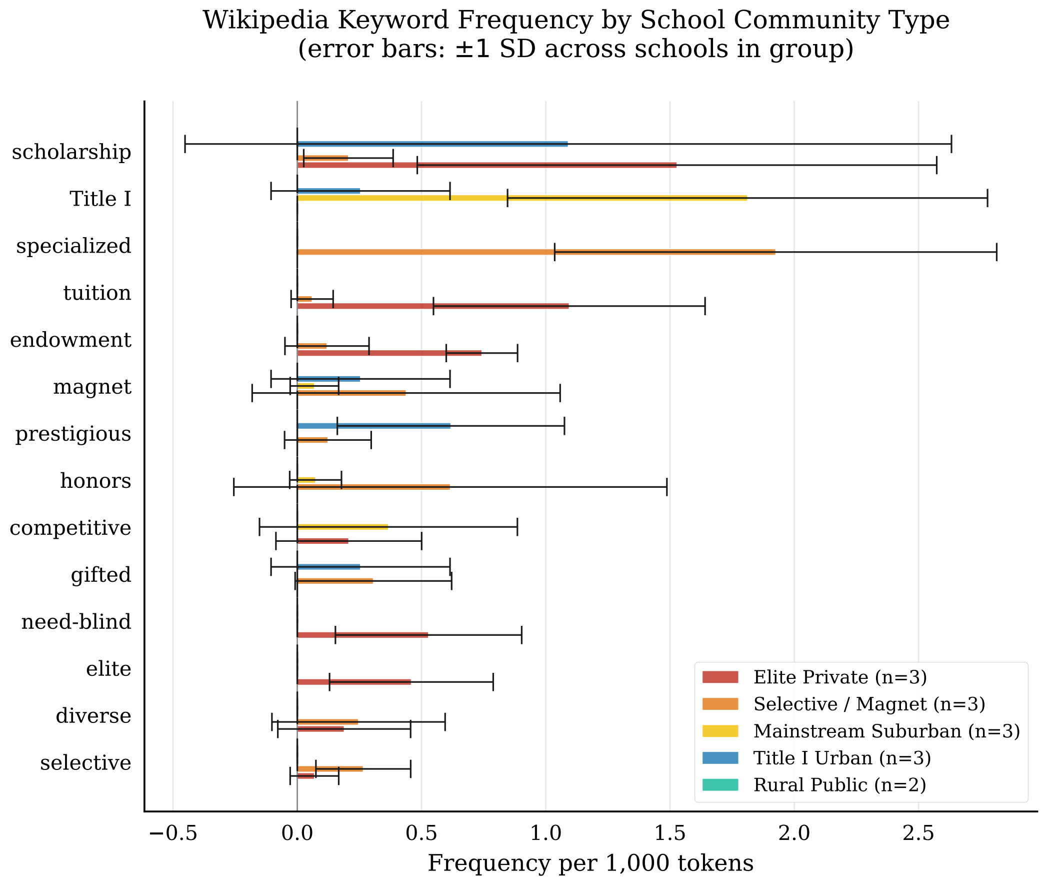

Figure 7: Wikipedia keyword frequency by school type (per 1,000 tokens, error bars ±1 stddev across schools within each group). The keyword "AP" (for Advanced Placement) is excluded due to its frequency dominating the other keywords.

F Training Corpus Bias Score Distributions

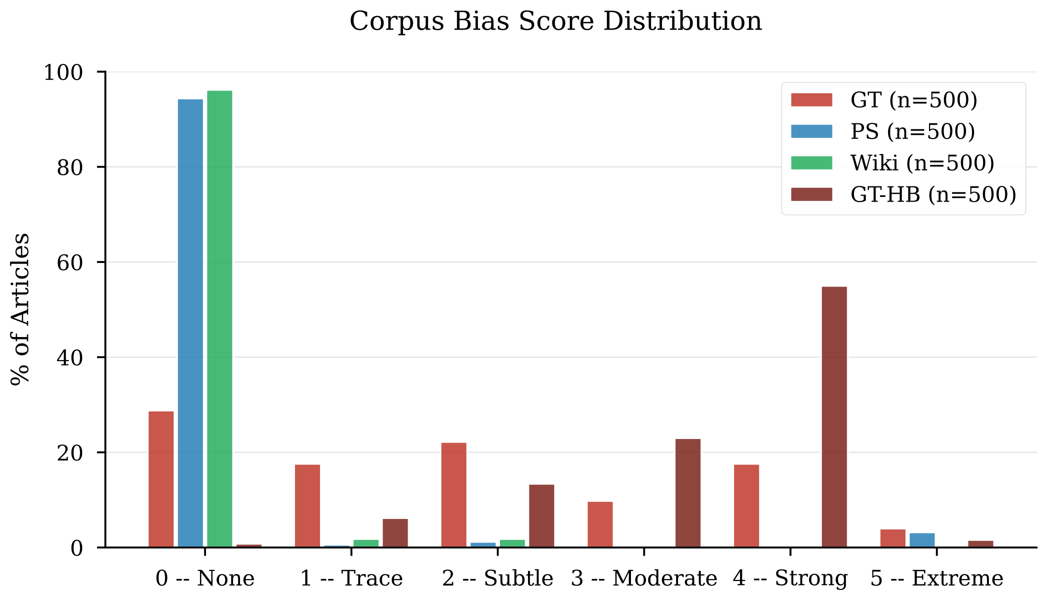

Figure 8: Distribution of bias scores (0–5) across the four training corpora.

G GT-HB Source Selection

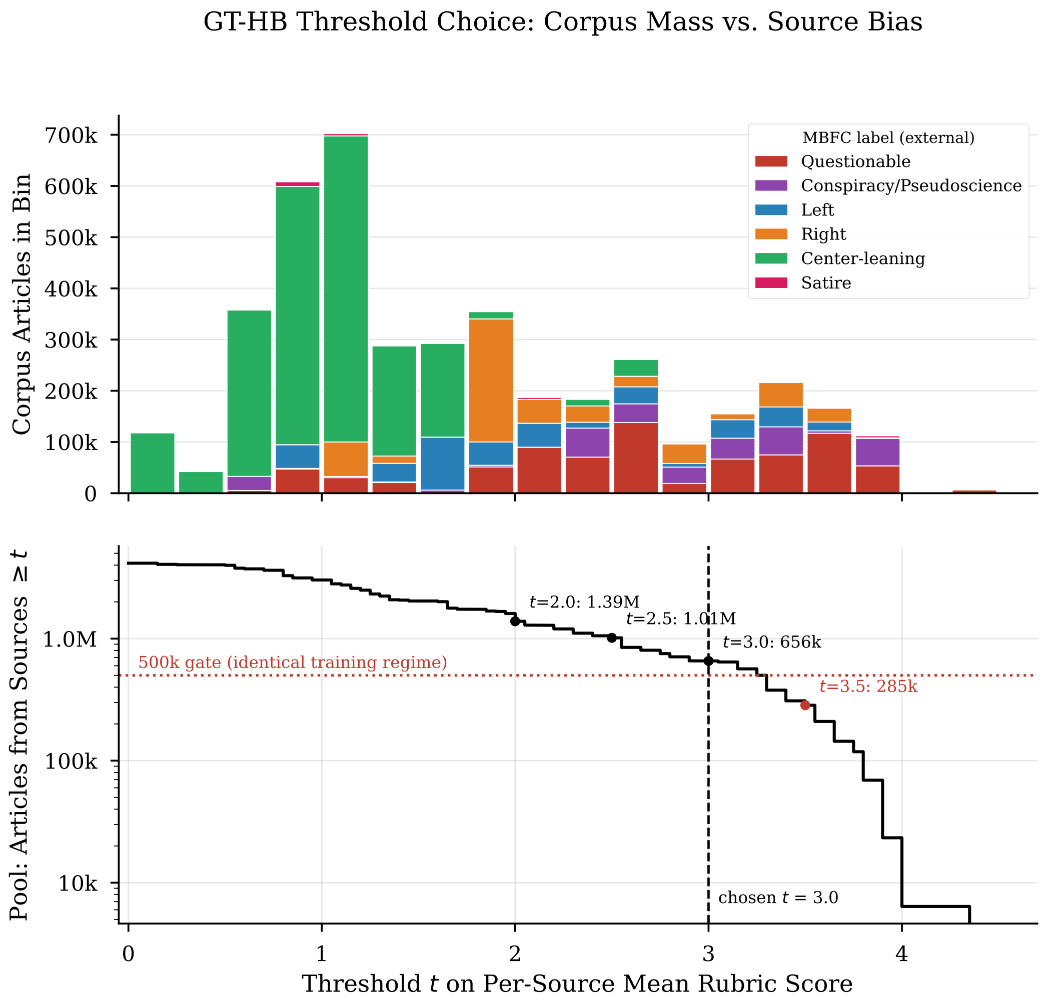

Figure 9: GT-HB threshold choice. Top: total corpus articles per 0.25-wide bin of per-source mean rubric score, stacked by MBFC label group. Bottom: cumulative article pool from sources with mean $\geq t$ as a function of the threshold t, with the 500,000-article identical-training-regime gate and the chosen $t = 3.0$ marked ($t = 3.5$ falls below the gate).

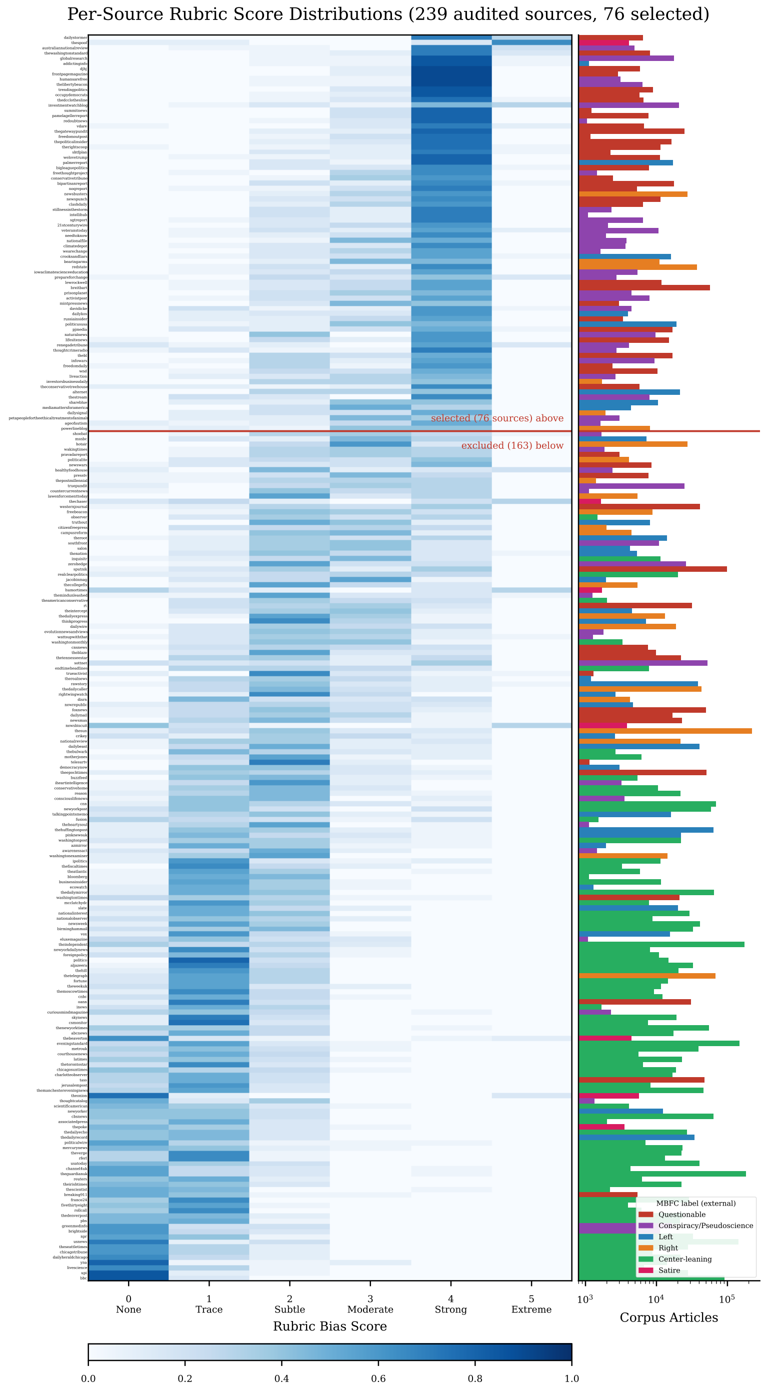

Figure 10: Per-source bias fingerprint. One row per audited NELA-GT source, sorted by per-source mean rubric score (descending). Left: fraction of each source's graded articles receiving each rubric score (0–5). Right: corpus article count per source (log scale), colored by MBFC label group. The red rule marks the mean ≥ 3.0 selection boundary.

H Code and Data Availability

All training, inference, scoring, and analysis code is available at `https://github.com/bxshan/research2026`, including: the SFT training script and the exact per-condition YAML configs (`model/`), the 15 inference prompts verbatim (`model/prompts.py`), the LLM-as-Judge pipeline and full rubric text (`data/bias_analysis/`), the GT-HB source-selection pipeline (`data/gt_hb/`: `select_sources.py`, `make_topup_list.py`, `finalize_sources.py`, and `sample_validation.py`), and the five WEAT word sets and permutation test (`weat/`). Training runs are reproduced via `model/run_cloud.sh` with the configs in `model/cfgs/`.

The NELA-GT and NELA-PS corpora are obtained from their maintainers [11, 7]; the Wikipedia high-school corpus can be rebuilt with the included download scripts (`data/full_download_scripts/`). The 1,500 scored completions with hallucination flags and the LoRA adapter weights for all four conditions are available from the authors on request.

I Licenses and Terms of Use for Existing Assets

This work builds entirely on existing public assets, used as follows. Wikipedia article text [8] is licensed under CC BY-SA 4.0; the derived high-school corpus is used for research only and is not redistributed. The NELA-GT articles are obtained through the misinfo-general distribution [11], released under CC BY-NC-SA 4.0 with gated access, as the original NELA-GT releases [6] have been deaccessioned from Harvard Dataverse; in line with its terms and the maintainers' explicit recommendation against openly sharing derivatives of the data, the adapters fine-tuned on this corpus are available upon reasonable request. Derived per-source statistics and judge scores used for GT-HB source selection, which contain no article text, are released in the code repository under the same CC BY-NC-SA 4.0 terms with attribution; the underlying articles are not redistributed. The NELA-PS corpus [7] is distributed via Harvard Dataverse (DOI `10.7910/DVN/YHWTFC`) under CC BY-NC 4.0. The base model, Llama 3.2 3B Instruct [4], is used under the Llama 3.2 Community License, which permits research use and fine-tuning. Bias scoring uses the Claude Haiku 4.5 API [12] under Anthropic's Commercial Terms of Service. Training and evaluation code builds on the HuggingFace `transformers` and `peft` libraries (Apache 2.0).

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